

13 Pollack in the North Sea, Skagerrak, and Kattegat

pol.27.3a4 – *Pollachius pollachius* in Subarea 4 and Division 3.a

13.1 General biology

The existing knowledge of pollack biology is summarised in the Stock Annex. According to this information it is a benthopelagic species and is found down to 200 m depth. In Skagerrak, 0-group pollack are regularly found in shallow areas close to the shore. Pollack are therefore protected from the fisheries in the early life stages. Pollack move gradually away from the coast into deeper waters as they grow.

Spawning takes place from January to May, depending on the area, and mostly at 100 m depth. FAO reports maximum length at 130 cm and maximum weight at 18.1 kg. Female length-at-maturity is estimated at >35 cm, at 3–4 years of age and growth after age 3 is about 7 cm per year (Heino *et al.*, 2012). Pollack feeds mainly on fish, and incidentally on crustaceans and cephalopods.

13.2 Stock identity and possible assessment areas

WGNEW (ICES, 2012a) proposed, based on a pragmatic approach, to distinguish three different stock units: the southern European Atlantic shelf (Bay of Biscay and Iberian Peninsula), the Celtic Seas, and the North Sea (including 7.d and 3.a). In the ICES advice, it was, however, decided to include 7.d Pollack in the Celtic Seas Ecoregion.

13.3 Management

For Subarea 4 and Division 3.a there are no formal TACs for pollack, but catches of pollack should be counted against the quota for some other species when caught in Norwegian waters south of 62°N. There is a Minimum Landing Size of 30 cm in European Member States (Council Regulation (EU) 850/1998). No explicit objective has been defined, no precautionary reference points have been proposed, and there is no management plan. Analytical assessments leading to fisheries advice have never been carried out for pollack.

13.4 Fisheries data

Landings statistics for pollack are available from ICES, but are clearly incomplete in earlier years. From 1977 onwards, the data series appears to be reasonably consistent and adequate for allocating catches at least to ICES subareas. Considering that pollack is not subject to TAC regulations, a major incentive for mis- or underreporting is not present and landings figures are thus probably reflecting main trends in landings in the different areas.

Landings by country for the years 1977–2024 in Division 3.a (Skagerrak/Kattegat) and Subarea 4 (North Sea) are shown in Table 13.1. Figure 13.1 shows total landings in Subarea 4 and Division 3.a from 1977–2024. Two periods with high landings can be observed in the late 1970s and 1990s. In general, the total landings for both areas have declined, albeit more so in Division 3.a. Landings in Division 3.a have been fluctuating around a low level since 2000, with a further decline

in most recent years, with landings below 200 tonnes in 2021 and 2024. In Subarea 4, landings have been fluctuating between 1500 - 2000 tonnes since 2000 at consistently higher levels than in Division 3.a. Swedish fishers targeted pollack from the 1940s until mid-1980s, when landings sometimes amounted to over 1000 tonnes. From the 1980s, pollack started to decline severely and is today seldomly caught in the Kattegat or along the Swedish Skagerrak coast.

Nowadays, no fishery is targeting pollack, and it is mainly, possibly exclusively, a bycatch in various commercial fisheries. Norwegian catches peak in the months of March and April, and this may be associated with spawning aggregations. In 2024, 48% of the total landings were caught with gillnet and 48% with otter trawls in Division 3.a. In Subarea 4, 17% of the total landings were caught with gillnets and 67% with otter trawls. The geographical distribution of Norwegian otter trawl catches resembles those of the saithe fisheries, but the catches of pollack are much lower.

Pollack is rarely discarded. Discards are estimated at 1.58 tonnes in total for both Division 3a and Subarea 4 in 2024 (see Table 13.2 for total landings and Table 13.3 for estimated discards). In 2025, UK England updated its 2021 to 2023 data. Changes were minimal; much less than 1% change in all years; a decrease of 1 tonnes in UK England landings to 382 t in 2021, an increase of 2 tonnes of landings to 557 in 2022 and no change for 2023. Discards were raised for all nations. Raising was done using 2 groupings for Subarea 4; i) One grouping or all GNS gear in all seasons/year and ii) one “bucket grouping” for all remaining fleets. For Division 3a three groupings were used; i) One grouping for all OTB trawls in all seasons/year ii) one grouping for all GNS gear in all seasons/year and ii) one “bucket grouping” for all remaining fleets.

In 2024, below minimum size (BMS) landings and logbook reported discards were reported to ICES for pollack. No BMS landings or logbook reported discards were reported in InterCatch whereas 13 kg of BMS landings were recorded in the preliminary official landings.

Pollack is also frequently caught in recreational fisheries. Regularly collected data about these catches are not available to the working group. Norwegian recreational fisheries data collected in 2009 suggests that catches of pollack south of 62° north in the tourist fishery may range between 13–30 tonnes (Vølstad *et al.*, 2011), whereas a more recent study (Verter *et al.*, 2023) estimated all recreational catches from hook-and-line fisheries in western Norway from April 2018 to March 2019 to be as much as 66 tonnes in this area alone. Denmark reported an estimated 35 and 58 tonnes of recreational catches in 2022 and 2023, respectively.

13.5 Survey data/recruit series

Pollack is caught in small numbers during the IBTS surveys covering Division 3.a and Subarea 4 (Figure 13.2). Pollack is mainly distributed in the northern North Sea (along the Norwegian Deep) and in the Skagerrak-Kattegat. Time series of abundance (average number per hour) in the IBTS are shown for Subarea 4 and Division 3.a separately, for quarter 1 (from 1983 onwards) and quarter 3 (from 1991 onwards) (Figure 13.2). The catches are low, and rather irregular, and no clear patterns emerge in Division 3.a and Subarea 4. However, in both surveys, larger numbers per hour are observed in 2024.

13.6 Biological sampling

Preliminary analysis of length data in Subarea 4 and Division 3.a by Norway in the period 2010–2013 indicates that length ranges of pollack caught in gill net fisheries differ with mesh size and location. The majority of fish caught in western Norwegian fjords had a size range of 60–80 cm (Figure 13.3) compared to 50–70 cm in the Skagerrak (Figure 13.4). More recent length samples from commercial bottom trawl catches collected by UK-Scotland, Germany and France in

Subarea 4 in 2024 indicated that the majority of catches were within a size range of 50-80 cm (Figure 13.5), a size composition similar to previous years. Data from the Norwegian oceangoing reference fleet in 2023, similarly show landings of large individuals between 60-90 cm (Figure 13.7).

13.7 Analysis of stock trends

In previous years, the study by Cardinale *et al.* (2012), which analysed the spatial distribution and stock trends for the period 1906–2007 based on IBTS Q1 and commercial catches, was used to assess the stock for Division 3.a (Skagerrak and Kattegat) and it was found that there was a large decline in stock size from approximately 1960 to 2000. However, during routine IBTS surveys in Subarea 4 and Subarea 3, pollack catches seem rather irregular and with no clear pattern. A spatial analysis of Norwegian fisheries data from 2013 showing total Pollack catches by ICES rectangle, indicates that the surveys do not cover the geographic distribution of the species adequately in both Subarea 4 and Division 3.a (Figures 13.6 and 13.8). The surveys may therefore not be very well suited for monitoring this species as trends in standardised CPUE likely are not a reliable indicator for the status of the stock. However, if the stock increases, it is arguably expected that present trawl surveys (e.g. IBTS) would be able to detect such a stock trend in a consistent manner (Cardinale *et al.*, 2012).

13.8 Catch advice

In 2025, no new advice is calculated for this stock. In 2024, advice for 2025-2027 was calculated using the ICES framework for category 5 stocks (ICES, 2012b). For stocks without information on abundance or exploitation, ICES considers that a precautionary reduction of catches should be implemented where there is no ancillary information clearly indicating that the current level of exploitation is appropriate for the stock. The precautionary buffer was last applied in 2021 and therefore the precautionary buffer was applied in 2024. The average discard rate (2021-2023) was 0.185% of the total catch.

Previous catch advice (advised catch for 2022-2024)	1828 tonnes
Average discard rate (2021-2023)	0.185%
Precautionary buffer	Applied0.8
Catch advice*	1462 tonnes
Projected landings corresponding to catch advice**	1460 tonnes
% Advice change***	-20%

* Previous catch advice (advised catch for 2022-2024) × precautionary buffer

** Previous catch advice (advised catch for 2022-2024) × precautionary buffer × [1 – discard rate].

*** Advice value for 2025-2027 relative to the advice value for 2022-2024.

The current advice, 1462 tonnes, is now based on the previous advice (1828 tonnes), which in turn was calculated based on average catch rates and discard rates for 2018-2020 and the application of a 20 % precautionary cap (see also Tables 13.2 and 13.3). Landings in 2024 (2045 tonnes) overshot the advised catch for 2024 (1828 tonnes).

13.9 Benchmark issues list

13.9.1 Data

In order to get a better understanding of growth and maturity, WGNEW recommended that the collection of otoliths and maturity should be continued during the IBTS surveys for a few years. WGNSSK recommends also that the Norwegian biological data, e.g. age readings of pollack otoliths, from commercial catches should be processed. This effort is currently underway. Length information from commercial catches is also available from UK Scotland, Germany and France in InterCatch for the last ~5 years and the same information is being sought from Denmark and Norway. Potentially such data can be used to establish future reference points for this stock. Other surveys than IBTS are also currently being explored to evaluate their usefulness as potential indices for pollack stock size and/or recruitment. Potential candidates are for example the annual shrimp survey taking place in the Norwegian trench in January, and a preliminary index combining this survey with IBTS data is showing some promise.

13.9.2 Assessment

No assessment model exists for pollack. Once more data comes available for this stock, assessment methods could be investigated.

13.9.3 Forecast

There is no forecast for pollack.

13.10 References

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13.11 Tables and figures

Table 13.1. Pollack in Subarea 4 and Division 3.a. Landings (tonnes) by country as officially reported to ICES 1977–2024.

	ICES Division 3.a							Official Total
	Belgium	Denmark	Germany	Netherl.	Norway	Sweden	UK	
1977	10	1764	4	3	449	706		2936
1978	1	2077	4		556	794		3432
1979	13	1898	<0.5		824	1066		3801
1980	13	1860			987	1584	<0.5	4444
1981	5	1661			839	1187	1	3693
1982	1	1272			575	417	<0.5	2265
1983	2	972			438	288		1700
1984	2	930	<0.5		371	276		1579
1985	-	824	<0.5		350	356		1530
1986	4	759	<0.5		374	271		1408
1987	6	665			342	246		1259
1988	4	494			350	136		984
1989	3	554			313	152		1022
1990	8	1842	<0.5		246	253		2349
1991	2	1824			324	281		2431
1992	8	1228			391	320		1947
1993	6	1130	1		364	442		1943
1994	5	645	<0.5		276	238		1164
1995	10	497			322	271		1100
1996		680			309	273		1262
1997		364	<0.5		302	178		844
1998		299			330	105		734
1999		192			342	88		622
2000		199			268	33		500
2001		201	1		253	46		501
2002		228	3		202	44		477

ICES Division 3.a								
	Belgium	Denmark	Germany	Netherl.	Norway	Sweden	UK	Official Total
2003		168	3	1	236	17		425
2004		140	2	4	179	34		359
2005		160	5	7	173	37		382
2006		103	10	3	178	36		330
2007		172	9		245	38		464
2008		166	5		247	33		451
2009		208	7		220	38		473
2010		313	8	1	195	35		552
2011		193	7		168	28		395
2012		200	7		171	37		414
2013		210	3		172	35		420
2014		191	5	1	156	30		383
2015		190	14	1	138	48		390
2016		151	8	1	134	47		341
2017		185	7	4	117	44		357
2018		226	10	1	105	64		406
2019		196	5	1	81	30		313
2020		162	5	18	47	17		249
2021		131	2	1	40	17		192
2022		117	2	5	54	23		201
2023*		170	11	<0.5	47	18		246
2024*		139	5	<0.5	36	14		194

* Preliminary

Table 13.1. Pollack in Subarea 4 and Division 3.a. Landings (tonnes) by country as officially reported to ICES 1977–2024. (continued)

ICES Subarea 4											
Year	Belgium	Denmark	Faroes	France	Germany	Netherl.	Norway	Poland	Sweden	UK	Total
1977	121	275		75	142	38	419	9	0	442	1521
1978	102	249		98	154	21	492	2	0	471	1589
1979	62	333		72	64	8	563	11	31	429	1573
1980	82	407		66	58	2	1095		38	355	2103
1981	59	500		173	21	2	1261		12	362	2390
1982	46	431		59	40	1	1169	33	23	270	2072
1983	58	481		79	44	1	1081		57	300	2101
1984	52	402		108	37	0	880	2	106	315	1902
1985	14	308		69	23	0	686		51	363	1514
1986	44	550		45	21	0	602		67	362	1691
1987	21	427		988	21	0	471		40	290	2258
1988	32	432		367	30	10	560		20	296	1747
1989	31	273		0	21	4	568		37	269	1203
1990	44	924		0	34	3	651		126	366	2148
1991	31	1464		0	48	4	887		153	684	3271
1992	49	794		18	59	7	1051		141	1310	3429

ICES Subarea 4											
Year	Belgium	Denmark	Faroes	France	Germany	Netherl.	Norway	Poland	Sweden	UK	Total
1993	46	1161		8	161	19	1429		217	1561	4602
1994	42	635		12	55	14	845		113	872	2588
1995	56	532	1	7	84	18	1203		175	1525	3601
1996	13	366		4	99	13	909		82	945	2431
1997	20	272	1	1	115	11	733		82	1185	2420
1998	21	265		7	44	5	567		75	780	1764
1999	21	288		0	62	5	768		72	636	1852
2000	45	291		24	38	5	880		91	877	2251
2001	36	156		6	40	1	860		63	809	1971
2002	27	234		6	112	0	879		68	711	2037
2003	13	191		9	82	1	971		36	837	2140
2004	28	162		5	57	0	517		16	612	1397
2005	26	173		3	128	3	511		46	477	1367
2006	18	152		4	80	1	545		12	587	1399
2007	18	192		130	137	2	754		43	905	2181
2008	15	150		129	114	1	840		46	999	2294
2009	13	121	2	6	50	1	668		32	658	1551

ICES Subarea 4											
Year	Belgium	Denmark	Faroes	France	Germany	Netherl.	Norway	Poland	Sweden	UK	Total
2010	12	163		10	129	0	599		32	540	1485
2011	12	106	0	10	67	0	580	0	35	489	1299
2012	17	123	0	3	102	1	433		42	443	1164
2013	17	128	0	2	66	4	371	0	29	463	1080
2014	24	121		32	145	1	476		40	377	1215
2015	20	183		3	237	3	473		50	627	1594
2016	21	127		2	107	2	447		37	430	1174
2017	18	187		8	269	3	510		44	511	1551
2018	14	139		23	154	2	739		30	486	1588
2019	20	184		23	159	6	893		38	556	1879
2020	22	241		20	262	9	1125		71	615	2365
2021	15	136		60	181	8	636		75	382	1492
2022	8	160	0	35	193	3	615		75	561	1651
2023*	22	122		48	219	6	749		82	578	1826
2024*	15	158		85	151	11	751		91	583	1845

* Preliminary

Table 13.2. Pollack in Subarea 4 and Division 3.a. ICES landings (tonnes) by country for 2013–2024.

ICES Division 3.a												
	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Denmark	214	192	192	152	187	229	196	163	138	117	170	139
Germany	11	6	35	7	11	13	5	5	2	2	11	5
Netherlands	<0.5	0	0	1	5	2	1	18	1	6	<0.5	<0.5
Norway	174	156	138	135	117	108	81	48	40	54	47	36
Sweden	36	30	46	47	43	64	30	17	17	23	18	14
ICES Total	435	384	413	343	363	415	307	251	199	202	246	195
Official Total	420	383	389	338	357	406	314	251	192	201	246*	194*
Diff ICES-Off	15	1	24	5	6	9	-6	0	7	1	0	1

* Preliminary

ICES Subarea 4												
	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Belgium	17	24	20	21	18	14	20	22	14	9	22	15
Denmark	150	122	183	127	187	139	184	241	136	160	122	158
France	2	32	2	2	8	46	24	20	61	36	48	86
Germany	59	145	216	107	267	151	159	262	179	192	220	152
Netherland.	3	1	2	2	2	2	4	8	7	2	5	11
Norway	379	481	466	440	508	738	901	1129	636	605	749	751
Sweden	29	41	50	36	44	30	38	71	75	76	81	92
UK	456	377	626	423	508	488	569	621	383	557	583	585
ICES Total	1103	1227	1567	1159	1543	1608	1899**	2374	1492	1637	1831	1848
Official Total	1080	1215	1594	1174	1551	1588	1881	2373	1492	1651	1826*	1845*
Diff ICES-Off	23	12	-27	-15	-8	20	18	1	0	-14	5	3

* Preliminary

**Swedish catches for Subarea 4 were added manually to the data after exporting the data from InterCatch.

Table 13.3. Pollack in Subarea 4 and Division 3.a. ICES discards (tonnes) by country for 2013–2024.

ICES Division 3.a								
	Belgium	Denmark	Germany	Netherl.	Norway	Sweden	UK	Total
2013	0	1.949	0.139	0	1.795	1.528	0	5.41
2014	0	0.62	0.008	0	0.441	0.473	0	1.54
2015	0	2.026	0.385	0	0.667	0.094	0	3.17
2016	0	1.436	0.021	0.002	1.706	1.685	0	4.85
2017	0	1.152	0.047	0.001	0.892	0.237	0	2.32
2018	0	2.39	0	0	0	0.28	0	2.67
2019	0	0	0.856	0	0	0	0	0.86
2020	0	0.72	0.025	0	0.26	0.122	0	1.13
2021	0	6.67	0	0	0	0	0	6.67
2022	0	0	0	0	0	0.15	0	0.15
2023	0	0.096	0.012	0.0002	0.028	0.1	0	0.24
2024	0	0.27	0	0	0	0	0	0.27

[illegible]

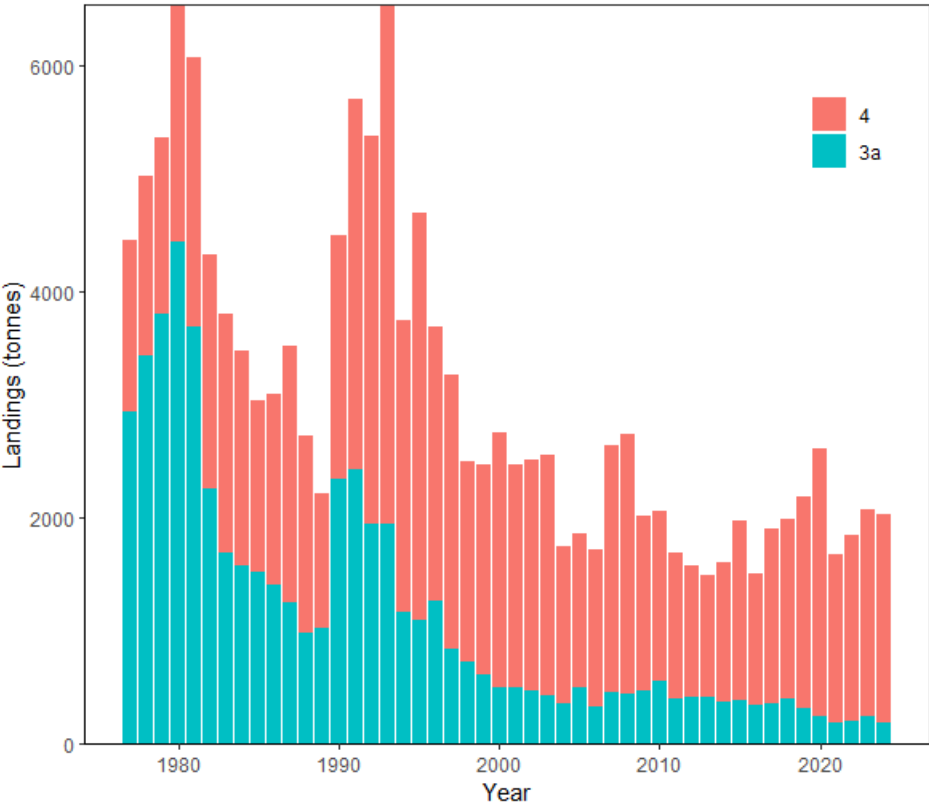


Figure 13.1. Pollack in Subarea 4 and Division 3.a. Total landings of pollack from 1977–2024 in Division 3.a and Subarea 4 as officially reported to ICES.

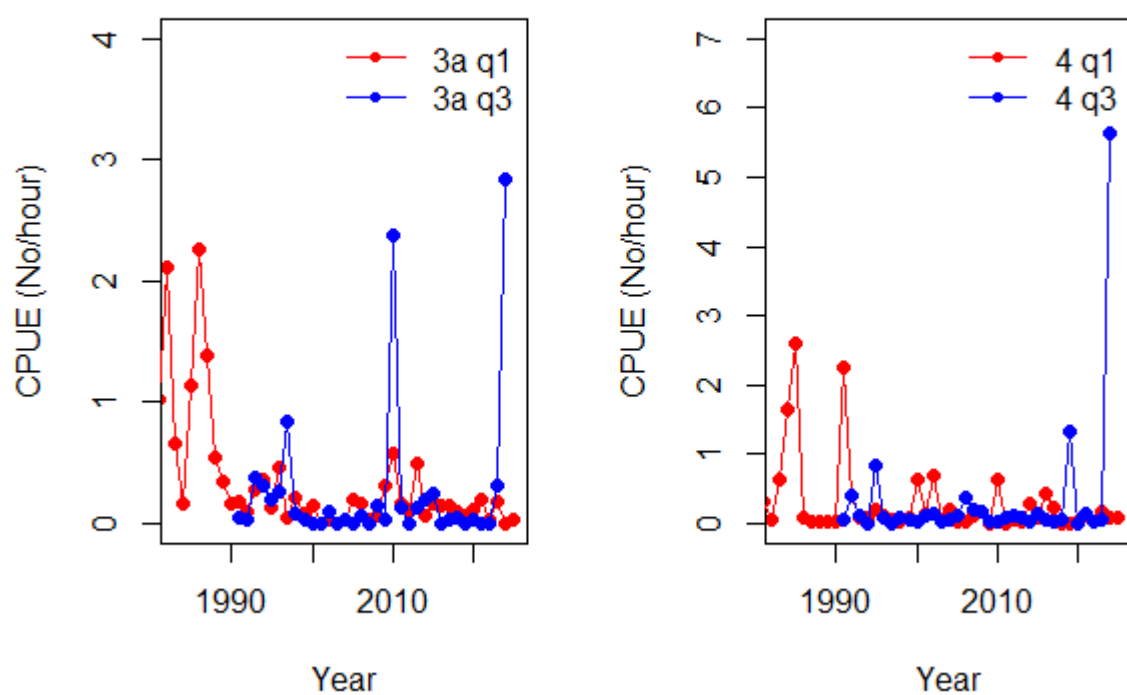


Figure 13.2. Pollack in Subarea 4 and Division 3.a. Catch per unit of effort (N/h) from IBTS surveys (GOV trawl, data from DATRAS) in ICES Division 3.a (left graph) and Subarea 4 (right graph) for the period 1983–2025 in Q1 (red) and for 1991–2024 in Q3 (blue).

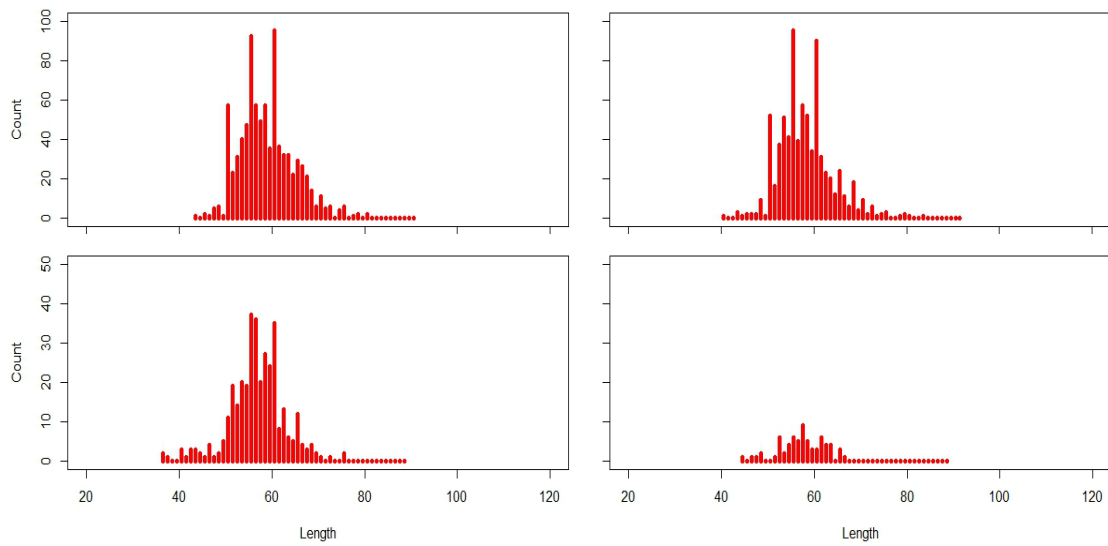


Figure 13.3. Pollack in Subarea 4 and Division 3.a. Length distributions of pollack sampled by the Norwegian reference fleet in the years 2010 (top left panel), 2011 (top right panel), 2012 (bottom left panel) and 2013 (bottom right panel) in Division 3.a. The data is aggregated for gillnets with a 63 mm mesh size.

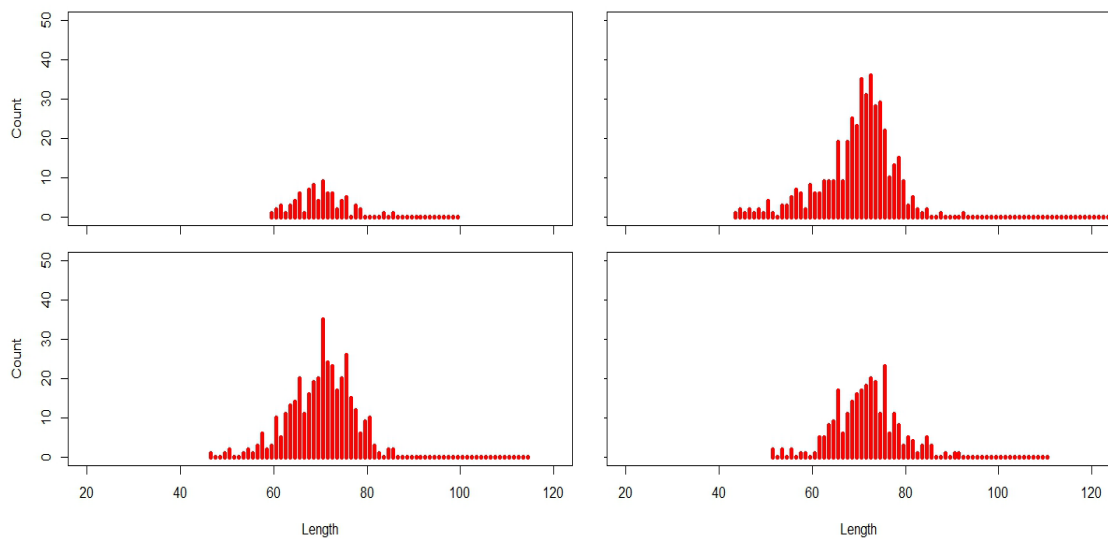


Figure 13.4. Pollack in Subarea 4 and Division 3.a. Length distributions of pollack sampled by the Norwegian reference fleet in the years 2010 (top left panel), 2011 (top right panel), 2012 (bottom left panel) and 2013 (bottom right panel) in Subarea 4. The data is aggregated for gillnets with a 70 mm mesh size.

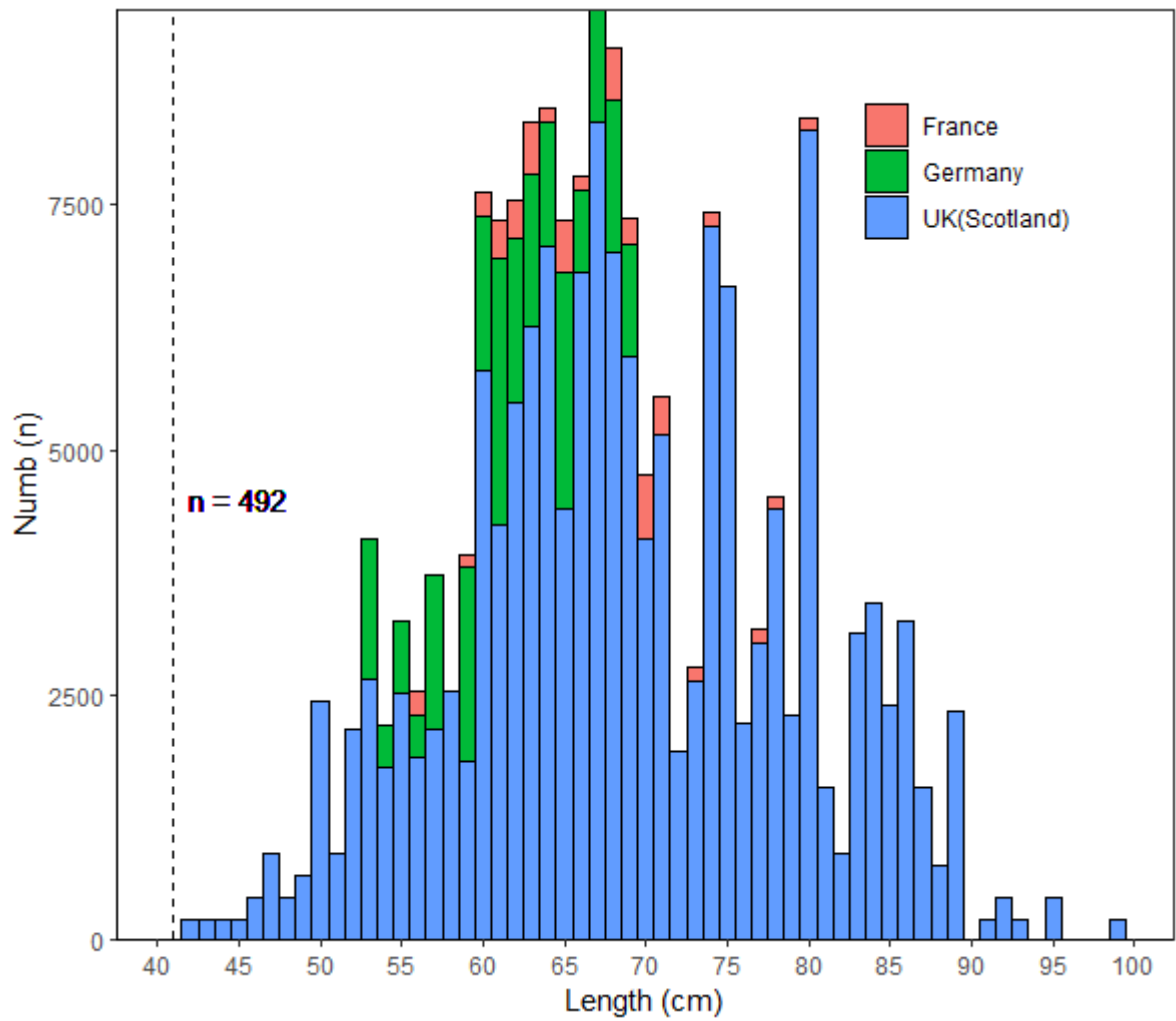


Figure 13.5. Pollack in Subarea 4 and Division 3.a. Length distributions of pollack sampled by commercial bottom trawls from France, Germany and UK (Scotland) in Subarea 4 in 2024. Figure shows samples only raised to fleet. Dotted vertical line indicates size at maturity.

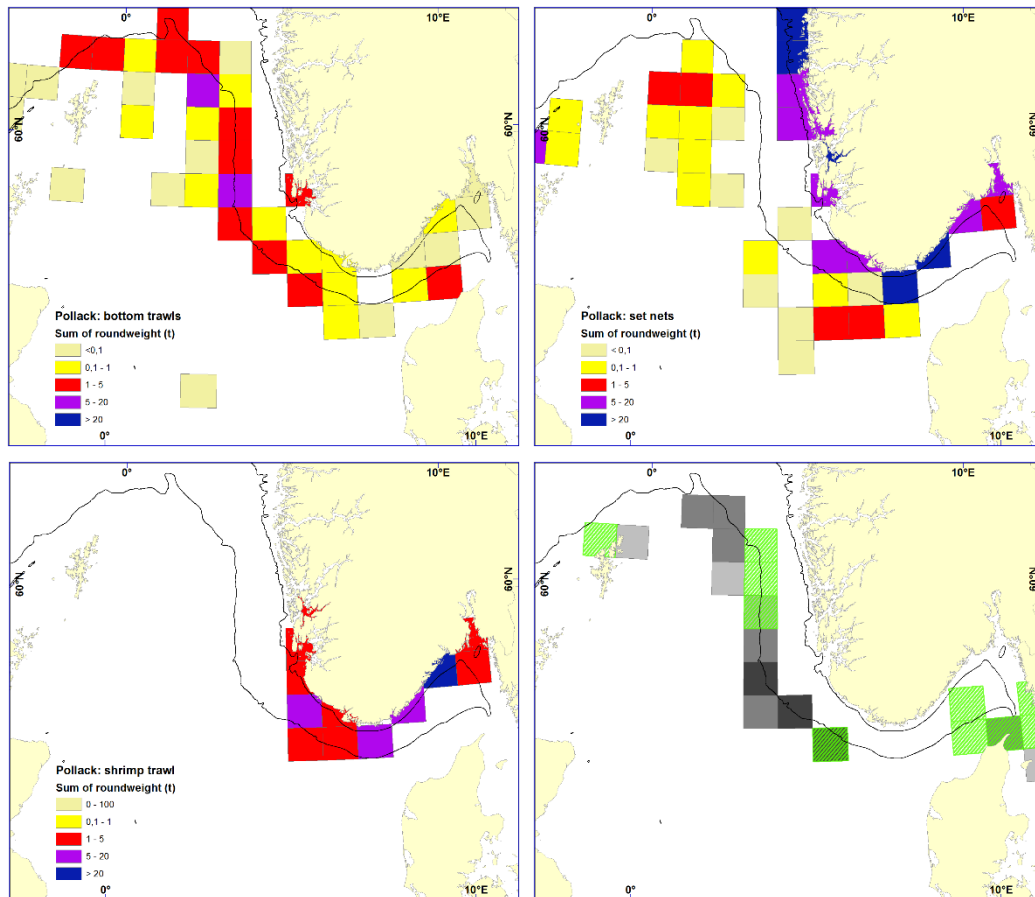


Figure 13.6. Pollack in Subarea 4 and Division 3.a. Distribution of Norwegian pollack landings in 2013 aggregated by fishing gear (bottom trawls, set nets, shrimp trawls), and pollack catches from IBTS surveys in 2012 (grey) and 2013 (green).

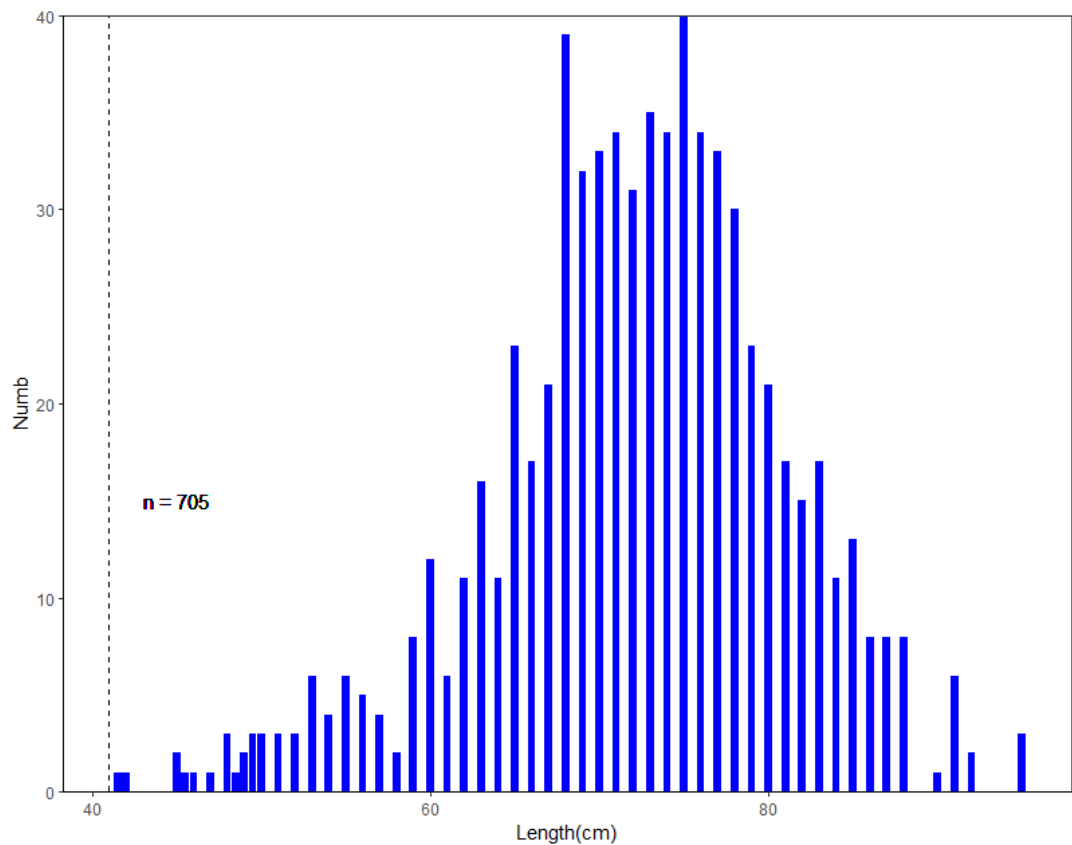


Figure 13.7. Pollack in Subarea 4 and Division 3.a. Length distributions of pollack sampled by the Norwegian oceangoing reference fleet in 2023. Figure shows samples. Dotted vertical line indicates size at maturity.

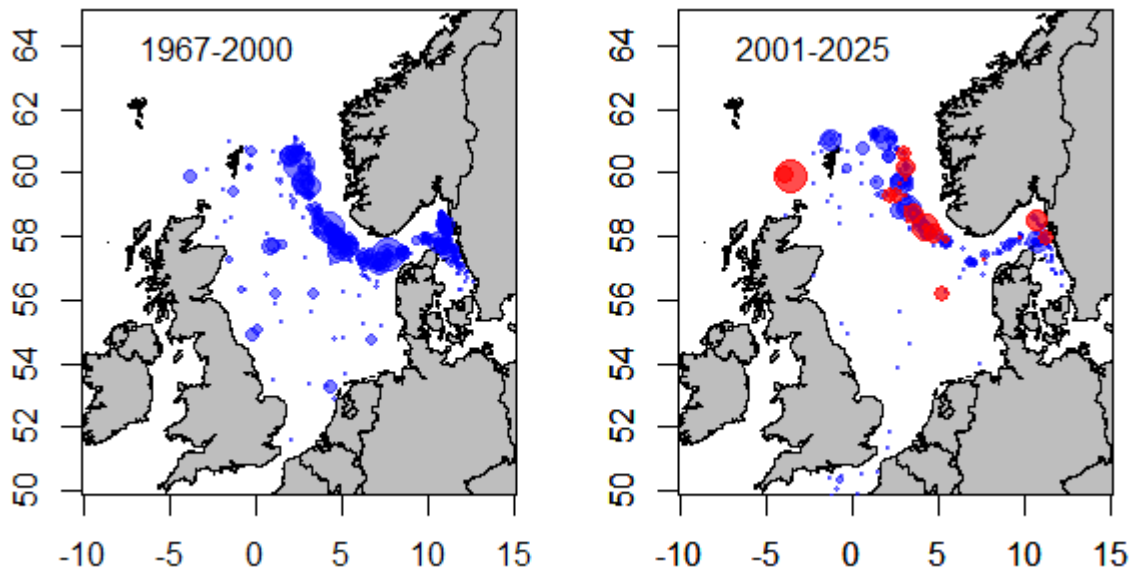


Figure 13.8. Pollack in Subarea 4 and Division 3.a. Pollack catches from IBTS Q1 and Q3 surveys 1967-2025. Left panel shows catches from 1967-2000 and right panel catches from 2001-2025. Catches from 2024 and 2025 are shown in red.

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